

Designerly Ways of Knowing – Soma Design Meets Industrial Design Thinking

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ABSTRACT

Soma design processes are known to be slow and thoughtful, demanding personal engagement, authenticity and deep somatic engagements with both ourselves and the interactive materials. What happens when such designerly ways of knowing meet with industrial design thinking? Together with IKEA – a large furniture manufacturer – we collaborated to design for better sleep. Based on years of experience and prior somatic knowledge of designing for body awareness, the academic team staged a progression of somatic experiences for a joint one-day workshop. We contribute to interaction design epistemology by uncovering frictions and alignments between two designerly ways of knowing with distinct identities and values. In particular, we critically reflect on how to approach material choices, somatic cultivation and the notion of solutionism.

CCS CONCEPTS

• **Human-centered computing → Empirical studies in interaction design**.

KEYWORDS

Soma design, interaction design, IKEA, design thinking

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1 INTRODUCTION

Soma design is an interaction design methodology intended to facilitate meaning-making and transformation in the long term [13]. It is a slow, demanding, first-person design stance that requires deep personal somatic engagements with interactive and sociotechnical materials. Soma design invites designers to *change*, becoming sensitive to their own bodily signs and signals and the potential aesthetics of the technological materials. We ask: *What happens when soma design ways of knowing meet with other designerly ways of knowing?* Through analysing a soma design process done together with designers at IKEA, a large furniture manufacturing company, we unpack some of the epistemological frictions and alignments between these particular two designerly ways of knowing. We illustrate the encounters through concrete examples, revealing the underlying values of the two stances. In particular, we critically reflect on how we approach material choices, somatic cultivation and the notion of solutionism. Through unpacking assumptions, adjustments made, value clashes and unfolding over the overall product design process, we aim to contribute more generally to design epistemology, untangling and pinpointing what constitutes a *designing body* trained in a particular tradition.

2 RELATED WORK

In this section, we provide a brief introduction to the design values and practices at IKEA and soma design.

2.1 IKEA values and design approach

As one of the largest multi-industry companies offering home furniture, design, and comfort at affordable prices to people in over 50 countries, IKEA has a vision of creating a better everyday life. Their stance has been framed as representing Swedish values of equality and an affordable good life for everyone [14]. The following are some of the key values IKEA identify with, infusing their design process:

Quality and affordability. IKEA's vision is shaped by a set of values summarised in the company motto, "*designing for the many people*", which translates into making accessible and good-quality furniture that blue-collar workers can afford. The values promoted

by the company are described in "*The Testament of a Furniture Dealer*" by IKEA founder Ingvar Kamprad [15]. In this document, he conceived IKEA as a democratising project, aiming to offer "*a wide range of well-designed, functional home furnishing products at prices so low that as many people as possible will be able to afford them*" without compromising functionality or technical quality.

Frugality. Another characteristic of Kamprad's vision was his open rejection of extravagance and wastefulness. IKEA embraces simplicity through its functional design, low production costs, and values infusing the company: *"We do not need fancy cars, posh titles, tailor-made uniforms or other status symbols."* Frugality is a guiding principle in IKEA's attention to the circularity in their design processes [1], which focuses on avoiding wastefulness while extending the life of products and materials. To do so, they engage in a process that includes four loops: reuse, refurbishment, remanufacturing, and recycling. Despite these strategies, IKEA's focus on affordability and large-scale production has not been free from criticism, being linked to the promotion of overconsumption, which contradicts its environmentally-friendly discourse [6].

Alignment with democratic values. The same Scandinavian tradition gave shape and form to Participatory Design [32]. Originally, participatory design came out of the co-determination law. The social democrats in the Scandinavian countries wanted to give workers power and influence over the tools they had to use as part of their work. The whole social democrat movement towards an egalitarian society for all was framed under concepts such as *Folkhemmet* – a home for everyone, providing good living and working conditions, rights to education, rights to leisure time, holidays, and good elderly care. A saying at the time was that the social democrats built the Folkhemmet and IKEA furnished it. That is, by providing affordable furniture for many people, everyone would be able to furnish their homes. To embed those values into their products, IKEA uses a human-centred design process, which they describe as a *value chain*, where democracy lies at the core of their methods [2]. The design process typically starts with *listening and learning*, where design concepts are sketched out and discussed closely with their suppliers. The following steps are *design and create, manufacture and improve, package and distribute* and *inspire and sell*.

2.2 Soma design values and methods

Soma design is a methodology and attitude that foregrounds the lived body, movement and subjective experiences as pivotal in the design process. It centres on cultivating somatic sensibilities towards the self, others and sociotechnical materials around us, influencing the design of meaningful multisensory experiences [13]. Soma design carries a strong set of values grounded in somaesthetics, a meliorative project coined by the pragmatist Richard Shusterman [28], who frames experience as an act grounded in the soma – a non-dualistic, self-agentic take on the body, proposing an active stance towards aesthetics – or our capacity to appreciate the world in which we are situated through all our senses. Below, we reflect on the main soma design values and how its methods carry part of the values we promote through our design process and outcomes.

Design through first-person. Soma design puts the subjective experience of the designer at the core of to the design process. Methods such as body maps [3], Feldenkrais sessions [7] or slow walks [5] are all intended as sensitisation tactics to ground designers' experience in their somatic self-awareness.

Feel more, speak less. Soma design promotes an attitude of experimentation and direct sensory engagement with technologies while delaying the rationalisation of phenomena. In concrete terms, that means that designers are exposed to different materials, technologies, and methods involving sensing and moving at the early stages of the design process [35, 37]. "*Feel more, speak less*" is presented as both a provocation and an invitation to suspend our usual ways of doing design and research. Methods, such as A Labs [1] and the soma bits toolkit [37] occur early in the design process, letting designers feel, touch and explore interactive materials, such as vibration, heat and shape-changing surfaces.

Cultivating empathy. Despite its strong grounding in first-person experiences, the cultivation of empathy is central to the soma design process. By developing somaesthetic appreciation as a skill, we increase our ability to derive meaning from our intercorporeal interactions with others. As we learn to identify the nuances of the movements and gestures of people we know, we cultivate our capacity to comprehend their emotions and reactions, gradually connecting with their experiences [11]. Here, the notion of cultivation is important, as developing somatic sensibilities happens slowly and after repeated engagements. Methods such as contact improvisation [24] sensitise designers to others' movements and gestures.

Plurality of users and experiences. Soma design embraces the plurality of bodies and experiences. Approaches such as the cultivation of vulnerability [25] and felt ethics [9] are encouraged in soma design processes, functioning as acts of care and attentiveness towards our similarities and differences.

Non-solutionist. The somaesthetic view permeates the values of soma design as a methodology. As a result, soma design methods are slow and non-committal, rejecting the premature imposition of solutionist language [22] and purposely staying undecided as a strategy to delve deeper into the ideation process. The undecided is valued as a generative space and strategy to foster the process of cultivation and reflexivity. The use of tactics such as estrangement [34] and magic machines [2] takes the designer outside of their comfort zone, fostering the generation of novel ideas and insightful reflections.

3 DESIGN PROCESS

The design brief entailed envisioning a commercially viable product for better sleep. As IKEA sells bedroom furniture and smart lighting systems, they wanted to engage with the whole experience of winding down, going to bed, and falling asleep as an integrated routine that could take advantage of these products.

3.1 Why set up a one-day workshop?

The academic soma design team had certain assumptions about what would be viable to IKEA designers. We assumed that the

[1] See their Circular Product Design Guide https://www.ikea.com/global/en/our-business/people-planet/designing-for-a-circular-future/

[2] As described in https://www.ikea.com/global/en/our-business/how-we-work/the-ikea-value-chain/

meandering, body-based processes, where we slowly gain somatic design sensitivities of both the design challenge (falling asleep) and the potential technologies that could assist that process, could not be shared in its entirety. Instead, to facilitate and speed up the soma design process, making it viable to share with our partners at IKEA, we set up a one-day workshop where we staged a tight progression of somatic methods and experiences, offering glimpses into our prior somatic experiences of designing for and with body awareness [12, 29, 37] By bringing our prior somatic expertise alongside semi-readymade interactive materials [35, 37], including heat elements or shape-shifting props, we were able to bypass part of the slowness characterising soma design processes. Instead, through our careful orchestration of workshop activities, we hoped that the IKEA team could, in only one day, come to somatically experience, appreciate and shape design ideas together with us.

As we will see, our assumptions that the IKEA designers did not have any deep somatic experiences of the design challenge were, in part, wrong. The IKEA teams' way of working already shared many aspects of soma design processes; they had, for example, all tried out various technologies as part of their everyday lives to explore what falling asleep might mean and which products already existed on the market. However, as we will see, certain aspects of our methods were novel to them, such as exploring somatic experiences in the same room together, articulating and sharing those experiences, and feeling and shaping interactions with technology already from the start.

3.2 Before the workshop – preparations in the academic team

The workshop took place in December 2022, but the roots of the design work started already in 2013. In 2013, an academic team of soma designers created the Soma Mat and Breathing Light for IKEA [31]. Many years later, in a long-term user study, it was shown that these designs indeed had long-term, transformative effects on its users' body awareness [30]. Following those early design experiments in 2013, soma designers engaged with many different domains, creating soma designs for everything from awareness of menstruation cycles to breathing exercises [4, 16, 18, 33]. In particular, the exploration of different breathing techniques has strongly informed our current approach, including exercises for singers to strengthen their muscles [18]; for relaxation, [30, 31]; and activation [17]. After learning certain breathing forms, such as the calming HRV-breathing pattern [19], we 'somatically knew' it would be important to explore in the IKEA collaboration. We also knew that certain Feldenkrais lessons would make sense, such as *Glueing the Lungs* – an exercise where you learn to extend your lungs in all directions, thereby deepening your breathing awareness. In our prior design processes, materials such as shape-shifting pillows or McGibbens muscles had been used in contexts related to the design challenge of falling asleep [16, 18]. Based on our designs, we constructed a toolkit with simple bits and pieces from these designs, named Soma Bits [36]. To the workshop, we brought a system where you designers can enact a simple breathing rhythm –such as HRV-breathing– through four pneumatic-driven pillows.

The design work done over the years has helped make us more sensitive to different ways of breathing, ways of engaging with haptic technologies, and ways of shaping interactions – an embodied way of knowing. In that sense, we brought our own sensitised selves – our trained designerly *somas* – to the workshop.

3.3 Before the workshop – preparations in the IKEA team

Likewise, the IKEA team had spent time exploring technologies for falling asleep. They had, for example, used meditation apps to see whether those would have any effect on their own behaviours. They had read up on breathing rhythms and explored them – mainly on their own, as an individual activity, often in their homes as part of their everyday activities.

The IKEA designers also brought deep knowledge of digital and physical materials. One of the designers was an engineer trained in how to create mobile apps and IoT products. Another was an industrial designer with years of experience in design concepts harmonising with the IKEA values. A third designer had specialised in designing with light. The IKEA designers also brought their extensive knowledge of products that IKEA already has on the market and how those could be adjusted to fit the design challenge. For example, IKEA has a range of light bulbs that can produce different forms of light and colours.

3.4 Onboarding Session

We organised several meetings to bring the two teams closer before the workshop. The main onboarding session was conducted as a video conference a week before the one-day workshop. To learn more about sleep, we invited a stress research expert, emeritus professor Torbjörn Åkerstedt, the former head of the Stress Research Centre at Stockholm University, to give a lecture. In his lecture, he discussed sleep hygiene, the hormonal cycles of the 24-hour circadian rhythm, and the importance of habits and rituals.

3.5 Workshop

The one-day workshop was held at IKEA's headquarters in Älmhult, Sweden, one week after the remote onboarding session. It was attended by: seven IKEA designers, five researchers, one master student from KTH, and three designers from a consultancy firm *It's Already Tomorrow*, acquainted with the soma design methods. Participants were encouraged to wear comfortable clothes, leave their shoes outside the room, and come with an open mind to create a common understanding of the felt sensations that might arise. All attendees signed a consent form describing the scope and goals of the activities, the data collection, and the following storage. Some captures of the workshop can be found in Figure 1

Glueing the Lungs. Our first main activity was a one-hour-long Feldenkrais exercise aimed at lengthening both sides of the lungs. The purpose was to make our attention travel inwards, towards our breathing system, becoming sensitised to the size and elongation of the lungs, how they are attached to bone and muscle structures, and how the fascia keeps everything connected. The exercise helped everyone to slow down and start noticing and feeling what relaxation and attention to the breathing system could bring. Participants documented the changing bodily sensations on body maps before and after the exercise.

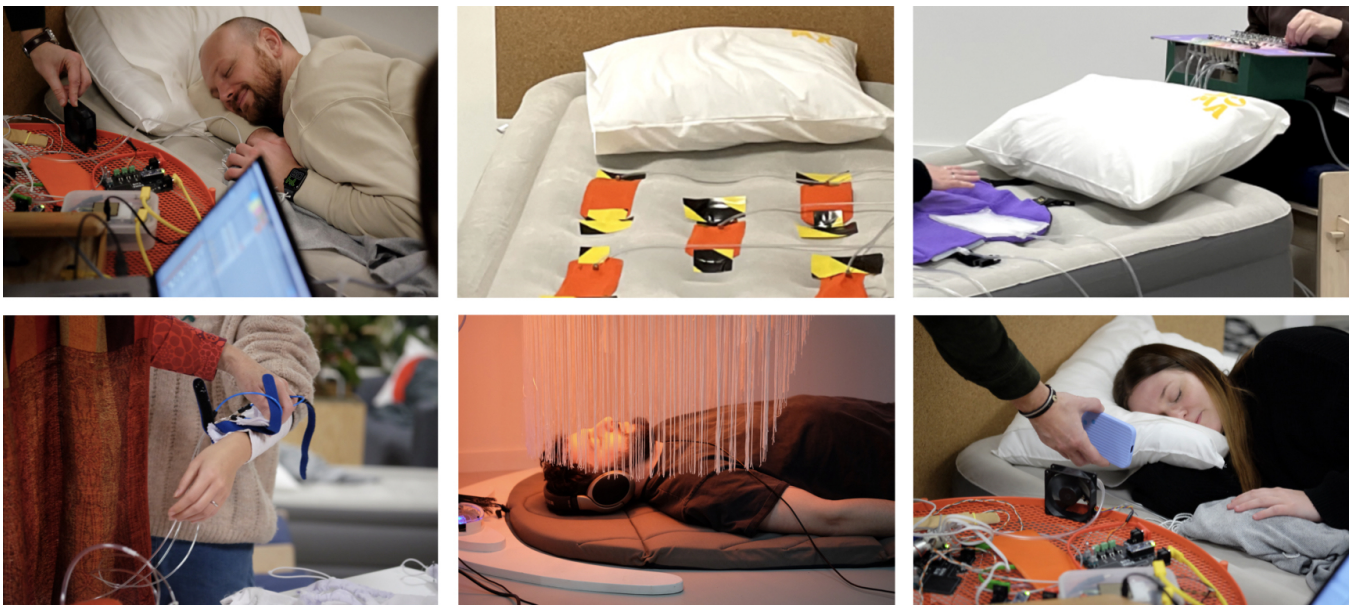


Figure 1: Design process.

Experiencing Pre-existing Designs. As mentioned above, based on prior work, four selected soma design prototypes were brought to the workshop that all participants were encouraged to experience. These were: (1) a pneumatics-driven HRV (Heart-Rate Variability) breathing system, (2) a shape-changing pneumatic-driven system that could be programmed to enact different inflation-deflation rhythms, (3) a system that generates light output in response to breathing and finally, (4) a mat with in-built heating pads that gently warms up, designed to bring awareness to different body parts.

Exploring with Digital Props and Other Materials. After lunch, we divided into four smaller groups. Each group was asked to prototype different concepts, building them from combinations of vibrotactile and heating props, and a curated selection of low-fidelity prototypes, lights, fans, and smartphones to play around with sound. The groups were encouraged to feel rather than talk about possible interactive experiences. Each group was given their own space with an air mattress, pillow, and blanket mimicking an actual bedroom, adding authenticity to the experience.

Devising a Design. After freely exploring for about an hour, each group was asked to create a dream scenario, recording a one-minute video to showcase for the rest of the groups afterwards. Inspired by the Magic Machines method [2], designers were encouraged to devise novel and imaginative ideas without feasibility limitations.

Sharing Ideas. The somatic interview and presentation that followed opened the design space for further discussion, articulation of felt sensations, and identification of shared and contrasting experiences.

3.6 Design concepts

Let us briefly discuss three of the design concepts from the teams – see figure 2 for illustrations of these and other design concepts devised in the one-day workshop:

- *The Living Blanket* was imagined as a soft, huggable blanket that inflates and deflates in a gentle breathing rhythm. The group emphasised that: "*It is really important to have all the modalities in total sync for the experience to be optimal.*" They suggested making the product compatible with a range of IKEA Home Smart products, allowing users to customise their experience and gradually introduce complexity.
- The Falling Asleep Routine was a comprehensive routine that commenced 30 minutes before bedtime and ended with falling asleep. The team stressed the significance of changing habits and creating a full routine encompassing the entire household. To establish non-intrusive communication between smart devices and users, the group proposed using lights as subtle cues to signal the start of the winding down process; "*Using lights to subtly communicate with the users, nudging them to start preparing for bed in a non-intrusive way. If the light pulsates in a desired breathing pattern, it can help guide the user to find a relaxing rhythm.*" This process would begin in the living room and continue in the bathroom while brushing your teeth. As the user laid their head on the pillow, the bedroom light would gradually dim darker and darker and finally be turned off, while a breathing rhythm would be maintained through an inflating/deflating pillow.
- *Deflation Shape Change* made use of shape-changing pads on a mat placed underneath the participant's back while lying on an air mattress. The shape-changing pads inflated and deflated in a breathing rhythm following the participant's breathing.

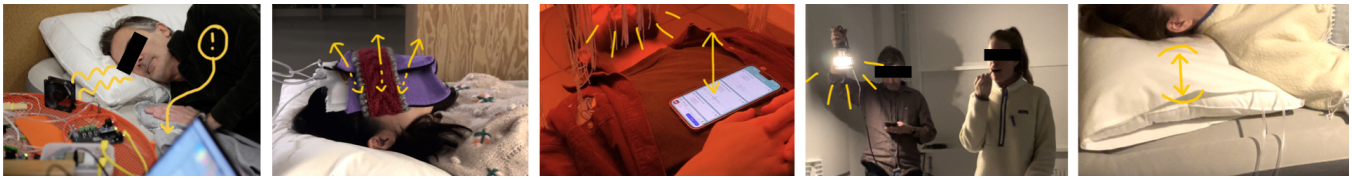


Figure 2: Design concepts. The first photo shows the Living Blanket. The second photo shows a shape-changing sleeping mask covering the face. The third photo shows a dimming light controlled by a mobile phone. The fourth photo shows the beginning of the Falling Asleep Routine, brushing teeth, where the light is being dimmed in the room. The fifth photo shows a pillow moving up and down in a slow breathing rhythm, a variant of the Deflation Shape Change.

Creating a ritual and maintaining consistency in late-night habits was echoed in the final discussions. One participant reported, "*I was stuck on something that Professor Torbjörn Åkerstedt kept repeating, that it is all about finding rituals for going to bed, rather than the moment you fall asleep – it's a journey, and we need to design for that*".

3.7 What Soma Design Brought to the Table

As argued above, the workshop successfully took all of us through a process of sensitising our somas, both turning inwards to feel our ability to breathe, slowing down our breathing patterns, extending breathing, as well as turning towards felt experiences of modalities and technologies, such as vibration, shape-change, heat, sound and light. Some important, somaesthetically grounded, felt insights grew out of this engagement. Here, we will focus on the insights and exploration methods that were novel to the IKEA design team – bringing insights that their prior design explorations had not yet delivered.

Substantial individual differences. When participants compared their experiences, they found substantial individual differences –beyond what they had assumed. Some were sensitive or disturbed by certain modalities, others not. Some needed more extended rituals to wind down, while for others, shorter rituals were enough. Some needed a slower breathing rhythm than others. While it may seem obvious that there will be individual differences in how we prefer to wind down and fall asleep, what was striking in the workshop were the noticeable differences in somatic experiences. Some spoke of the inner experience of their lungs, noticing the flow of air into the left or right lung, the muscles involved, and shifts in their belly, legs and shoulders. Others were unable to distinguish details of their breathing apparatus. Similarly, when experiencing the technology, other individual differences were revealed; for example, some found the motor sounds of the pneumatics soothing, and others disturbing.

To a soma designer, such individual differences are not unexpected and have been repeatedly documented in the literature describing soma design processes [29]. However, to a designer trained in user-centred design, the assumption is that most interactions that touch the body can be designed to fit many users by solely adjusting to physical differences and the ergonomics of different bodies. User-centred design also assumes that by engaging empathically with end-users, patterns in bodily responses can be found and met through design choices. The workshop made the connections between cognitive, emotional and bodily differences in experience clear.

Holistic, organic rhythm. The workshop also revealed nuances of experience that needed fine-tuning in order to come together as a whole. One experience quality that soma designers often emphasise is the overall rhythm of the experience [10]. One example of such holistic rhythm design concerned the HRV-breathing – the breathing pattern that attracted the most interest out of the four breathing rhythms we tried. Participants described it as producing a "*soothing sensation, like being on the water or in a hammock,*" and they appreciated the "*calm and consistent rhythm.*". However, the HRV breathing also revealed the importance of attending to nuances of the experience in order to find the right rhythm, pace, and intensity needed to arrive at a harmonious experience. The different modalities, shape-changing rhythm and the sound of the pneumatics were necessary to come together into a coherent experience. One participant commented "*The sound from the motor adds to the soothing experience; it reminds me of white noise,*" and that "*It guides the experience when going up and down, kind of like an inhale and exhale.*" The rhythm of the human breathing needed to be mirrored in the interaction.

But a simple mechanistic mirroring of a sinus rhythm is not enough. This became evident in the *Deflation Shape Change* prototype, where the inflatable pads were manually controlled by the two master's students from [anon], resulting in a slower process of air entering and exiting the mat. Participants expressed their appreciation for the feeling of being lifted and supported by the pillows, describing it as a "nice hug, making you feel tucked in" and feeling "embraced by it." The imperfection brought about by the manual pumping was mentioned as something favourable, with one participant noting that "It felt human because it wasn't all perfect, and I liked that!"

The holistic coming together of modalities to create a rhythm and organic feel often comes through in soma design as it so closely aligns with the organic bodily rhythms we engage with [13].

Devising rituals. To disrupt the habitual ways of going to bed, the groups employed methods such as bodystorming [23], defamiliarisation, and estrangement [34]. While the importance of a routine arose already in the lecture by Professor Åkerstedt, the enactment of what that routine could be, was enacted with those methods. In the final dream scenarios, two out of three groups specifically highlighted the importance of incorporating a routine.

Here, the IKEA designers appreciated the process of not solely imagining a ritual but enacting it together and sharing the experiences that arose.

Designing in situ. What was perhaps most unusual for the IKEA designers was the overall idea of enacting the whole scenario of falling asleep and sharing those experiences. As part of the soma design process, we try to design with bits and pieces of technologies in a setting relevant to the context we are designing for [13]. Designing for winding down and falling asleep, we argued, should, therefore, be done in a setting where participants could lie down and role-play falling asleep. This was a novel experience for the IKEA team; their style of designing would otherwise be to reason about a possible design, send the sketches to technicians and material experts, who would devise a prototype, send it back, and only then would they be able to experience what they had imagined. They appreciated the low threshold for trying out ideas: "*I really see the strength of low-fi prototypes since it allows everyone to try ideas quickly*". As one participant remarked, *"I was reminded of the fact that you don't always have to implement something to communicate the idea with developers, but rather invite them to the process and let them experience it"*.

While bodystorming is not uniquely a soma design method, the way it is enacted, with the emphasis on somatically experiencing the interaction, was novel to the IKEA designers.

Beyond the short experiences we can enact in a one-day workshop, a soma design process also asks you to live with the designs for a longer time period. This could obviously not happen in this one-day workshop, but was part of how we (the academic team) continued the work after the workshop.

3.8 Where soma design did not align

While the soma design processes brought important insights, rendering interesting ideas materialised in simple technologies, the most preferred and evocative design concept design concepts that arose in the one-day workshop were in conflict with core IKEA values.

Accessible materials and affordances. In the final reflection IKEA's interest and strong desire to ensure the accessibility of the end product was emphasised. One participant argued that "*It needs to be easy and approachable for most people,*" and "*Making something that gets people to improve their sleep quality by relatively small and affordable measures.*" Several of the design concepts from the four groups made use of shape-changing materials in combination with sound. As shape-changing materials are not yet on the market and definitely not affordable to many people, the IKEA designers quickly agreed that those could not be used in any final product, despite their evocative properties. This accessibility also relates to the importance of offering products that can be easily incorporated into users' everyday lives. As one participant pointed out, "*The challenge is how to get millions of people to actually experience something that gets them to create better rituals around sleep, without the privilege of having a bunch of experts around telling them what to do.*

While a soma design process can be restricted to working only with affordable materials, we brought to this workshop some materials, such as the shape-changing pads, which we later realised were too expensive and cumbersome to harmonise with the IKEA values.

Intimate experiences shared in a hierarchical company. Sharing the everyday, but still very intimate experiences of falling asleep puts participants in a potentially vulnerable position (cf. [26]). In the workshop, the IKEA participants belonged to different parts of the company and also different positions in the company hierarchies. Designing together always requires leaving our egos at the door, but soma design requires moving several steps beyond solely letting go of your ego. Soma design requires an authentic sharing of somatic experiences without the protection and distance to the self that verbalisation of an experience or design idea might bring. As mentioned above, the simple act of being asked to lie on the floor in a work setting is non-habitual, immediately breaking down existing hierarchies.

Continuity of design team over time. After the workshop, a subset of the design team was asked to bring the design ideas one step further, creating a fully functioning prototype tested in home settings to prove its value. This design team did not include any of the IKEA designers but was commissioned and paid for by IKEA. The team was asked to come to a compromise between what had been experienced in the workshop and the need for an affordable product for *the many people*. The team ruled out any shape-changing materials and instead based the design on pre-existing IKEA products – a smart speaker (TRÅDFRI) and the remote control of smart lights (TRÅDFRI). The final assembly comprised the SYMFONISK lamp base, a clear light bulb from the TRÅDFRI collection, a novel sandblasted glass shade with a gradual fading effect, and a 3D printed lock to safely attach the shade onto the lamp base; see Figure 3. The light pattern projects a particular breathing rhythm harmonised with an accompanying soundtrack. The sound alternates between two notes to mark the pace of the breathing pattern and has a spacious vibe that creates a sensation of losing track of time and space. After testing different breathing patterns, Resonance Breathing (HRV) with a 5-5 (inhale and exhale) tempo was chosen to be the most approachable technique, confirmed by feedback from the IKEA team who also tested the different breathing rhythms.

In a large company, there is no guarantee that the same designers will be working on the project for its whole duration. When the final design concept was defined and ready to be implemented, it turned out that the appointed developer at IKEA was new to the project and unfamiliar with soma design. Having not participated in the workshop, he lacked a prior frame of reference to build on. This crucial handover meeting was held in a room where the lights could not be controlled, the developer was pressed for time, and there was no opportunity to slowly experience the intended design concept. As a result, much time after that meeting was spent by the design team trying to convey the intended experience to the developer. Important nuances took time to fine-tune, a process that could have been much easier and quicker if everyone had collectively understood and experienced it together.



Figure 3: The final prototype: a glass-shaded lamp, casting shadows onto the ceiling, mounted on top of a loudspeaker.

4 FRICTIONS AND ALIGNMENTS IN DESIGNERLY WAYS OF KNOWING

Our joint one-day soma design workshop resulted in the sharing of bodily experiences and productive design outcomes. But the work also revealed differences in our *designerly ways of knowing*.

On the positive side, contrary to the resistance first person methods have been met with in the field of Human-Computer Interaction, the IKEA designers had no issues with engaging with their own experiences. As most designers working in industry, they had tested various commercial designs and ideas through first-person engagement [21], often in the privacy of their homes. What was novel to them was being asked to perform somatic, bodily acts together in the same room, sharing their experiences. The set-up with a room filled with yoga mats, beds, lamps and simple interactive technologies was not how they would typically work. While this initially created some insecurities about how to behave, it became the most appreciated aspect of the workshop. IKEA's open-mindedness towards soma design and alternative ways of working was genuinely evident. One of the UX designers was particularly satisfied by this way of working and mentioned that, despite his relatively long career, he had never encountered a similar process. He shared his interest in continuing to apply this methodology in other projects.

The individual differences that the process revealed, harmonised nicely with the IKEA values of *designing for the many people*. Designers at IKEA found somatically engaging with low-fi interactive materials particularly useful. While Schön speaks of *seeing-drawing-seeing* as the basic design process [27], soma designers speak of *shaping-feeling-shaping* [35] to describe the process of putting together experiences on or around the body. But *shaping-feeling-shaping* is hard to do when you do not have technological materials in the room. Putting together experiential prototypes on the fly, being able to *feel* them already then and there, was particularly appreciated. In the final reflections, the IKEA designers further reinforced their appreciation for this methodology, as they expressed how it allowed them to engage their bodies in the design process and quickly feel the different ideas in practice

While the commercial realities ruled our certain materials, we do not see this as particularly troubling to the underlying *designerly ways of knowing*. Nor do we find the frictions with hierarchical structures within the company or the unusual ways of engaging as a problem of epistemological stances. Those tensions are to be expected, and can be handled. Here, we would like to emphasise that this experience has shown us that in the same way, soma design values shape our design process, the IKEA team stand by their values to guide their work. The IKEA designers wanted to arrive at a design that would be useful and accessible, where people across the globe would get help in falling asleep. Their explorations prior to the workshop showed a deep personal engagement, informed by a strong company-driven value-based agenda. In fact, industry expertise and their rigorous standards of quality control can help achieve the vision of setting the stage for long-term, high-quality somatic experiences of design. It may give shape to soma-design's experiential understandings around somatic sensibilities involved in establishing rituals. Establishing rituals and novel habits is crucial to fostering a transformation process, and product robustness is instrumental to building a relationship of trust with the user.

What remains unclear is whether the simplified one-day introduction to a few bodily- and technological experiences led to the somatic knowledge and designerly sensitivity a soma design practice demands. Learning to notice small, fine-grained somatic signs and signals that may come with a particular breathing pattern or a specific light design, may take a long time to acquire. Furthermore, noticing is only the first step. Soma design asks us to not only examine but *improve* on connections among sensation, feeling, emotion, and subjective understanding and values. Being guided

through such a process is a way of sharing insight, but perhaps not a way of searching for those connections in yourself or learning how to change them.

In our prior work, we have found that those learning processes are slow and while a workshop can set the stage for an experience to more likely occur, the work involved in engaging and learning from that experience takes time. We often spend months shaping and reshaping designs as the process changes us and our understanding of what creates for deep somatics-based meaning making process. Once we invite end-users to experience our designs, their use also has to be slow, thoughtful, and active in order to shift from what is habitual to them into novel habits [30]. Changing your sleep routine is an excellent example of a shift from one habit to a novel one.

The workshop setup allowed us to share somatic knowledge, but by avoiding the meandering, non-solutionist processes of soma design, the process towards a designerly-grounded meaning-making process is, in parts, removed. In those processes, we challenge our bodily habits, attempt to move beyond preconceived design concepts, disentangle technologies from our somatics selves, and cultivate our designerly sensitivities [8, 20]. The slowness of soma design is not so much about being slow, but taking the space needed for this form of learning to take place.

However, designers at IKEA are not novices. Even if their somatic expertise does not manifest in ways that might seem as explicitly body-centric as in soma design, their craft involves a strong engagement with their bodies – particularly through their hands and all their senses as they interact, shape and try materials and prototypes. Even if our methods are different, designers who cultivate their expertise through and beyond digital materials –i.e. graphic, industrial, textile, illumination designers – are used to attuning to the nuances of sensoriality [20], going beyond the rationalised versions of design processes endorsed by mainstream HCI. Despite some initial hesitation, during the workshop we could witness how IKEA designers had no issues engaging in material explorations with us, seeing the value and meaning in our processes. In the end, soma design in many ways shares *ways of knowing* with form-giving design approaches, even if some values between industry and soma design might not always overlap.

Still, some noticeable differences exist in how we motivate our somatic approach to materials. Soma design rejects solutionism [12], which manifests in its slow and exploratory approach to cultivating and transforming. In academia, we understand design-making as a process of knowledge generation [38], which in soma design is strongly connected to cultivating somatic sensibilities. In industry, the design language is driven by solutionism, which limits exploration to shape fast and effective solutions within budget and profit margins.

This said, soma designers have a responsibility to help shape futures, proposing worlds we want to inhabit. Transforming our methods into viable design methods that industry will adopt is a means to impact people's lives. The fact that our collaborators are open to making space for these methods means they share part of these values.

5 CONCLUSIONS

The research contributes to unpacking epistemological similarities as well as differences in *designerly ways of knowing*. The crafts-side and first-person engagements were overlapping between IKEA- and soma designers in important manners. By juxtapositioning the soma-based designerly engagements with the IKEA designers' processes, we uncovered some differences. The strong emphasis on involving our whole bodies, integrating movement, feeling, understanding and values, and the demands that follow, were novel to our colleagues at IKEA. Experiencing and shaping concepts using low-fi interactive materials was another appreciated difference.

In unpacking this meeting, the aesthetic aims of the different *'designing bodies'* became clearer. These aesthetic aims pertain to particular values, particular forms of craftsmanship, particular design sensitivities, and particular ways of understanding future users.

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